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PHYSICIAN-INDUCED DEMAND FOR SURGERY

DISCUSSION PAPER

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ABSTRACT

Following up the earlier findings by Fuchs on surgeon-induced demand, this paper makes numerous data and econometric improvements in conducting a test of neoclassical and inducement theories. A simultaneous equation model is used to estimate physician demand and equilibrium fees for surgery from a sample of 350 PSUs over the 1969-76 period. The results provide definite support for the notion of competitive market failure--particularly in large metropolitan areas. Other things equal, fees and utilization are higher in surgeon-rich areas although our estimated shift elasticities were only about one-third those found by Fuchs. A statistically significant, albeit small, price elasticity of demand for surgery was also obtained, in contrast to Fuchs. Increasing monopoly and disequilibrium models are also tested without altering the basic findings.

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PHYSICIAN-INDUCED DEMAND FOR SURGERY

1. Introduction

Policymakers are increasingly concerned with the high costs attributable to unnecessary medical care utilization, particularly the performance of excessive amounts of surgery. Surgical operations of doubtful marginal utility drive up health care expenditures both through physicians' fees and through hospital charges. At best, such operations may be a misallocation of scarce health resources; at worst, they may endanger the health and well-being of patients who undergo them.

Per capita surgery rates have been rising steadily, over 30 percent from 1968-1977. Part of this increase can be attributed to improved access through Medicare and Medicaid [Bombardier *et al*. (1977)] and to the growing proportion of elderly persons in the population who undergo surgery at two times the rate of younger age cohorts [NCHS (1977)]. Nevertheless socio-demographic changes and improved health insurance coverage alone cannot account for this phenomenal rise in operations. Surgery rates have been increasing for all age groups, for example, and the rate of increase has actually accelerated somewhat from 1973-1977.

While technological advances and a changing age distribution are partly responsible for these disproportionate rates of increase, they do not explain the wide geographic variation observed for almost all surgical procedures. Lens extraction, to take just one example, is performed 65 percent more often in the West than in the South [NCHS (1973)]. Similar variations in surgical utilization also have been found across much smaller geographic areas by Lewis (1969), Wennberg and Gittelsohn (1973, 1975), and Roos *et al*. (1977). Moreover, no evidence was found in these studies that residents differed in their propensity to develop appendicitis, hemorrhoids, hernias, etc.

Variations in the incidence of surgery may be explained by variations in the surgeons themselves. England and Wales, with a nationalized health service, have a surgeon-population ratio only one-half that of the U.S. and perform one-half the number of operations with no discernible detriment in health status [Bunker (1970)].[1] Closer to home, surgeon-population ratios do vary substantially across geographic areas, and are positively correlated with the incidence of surgery [Bunker (1970); Lewis (1969); Wennberg and Gittlesohn (1975)].

Correlations do not necessarily imply causation, however, because of confounding demand-supply factors (e.g., insurance coverage). In a now classic article, Fuchs (1978) estimated a multi-equation model of surgical utilization and found very strong support for the hypothesis that surgeons induce more demand for their services. This paper builds on Fuchs' pioneering work by putting together a cross-section time-series micro-area data base with significant improvements in data scope and market area specificity. Due to data limitations, Fuchs was unable to construct a true price-of-surgery variable, using instead a proxy composed of office and hospital visit fees. The proxy did not discriminate between metropolitan and non-metropolitan areas either. Moreover, analysis was conducted only for 22 aggregated areas, 11 census divisions, urban and rural, for two disparate time periods, 1963 and 1970. For all these reasons, we have undertaken another, more disaggregated, look at the issue.

The next section summarizes the orthodox, neoclassical, and the inducement models. This is followed by an empirical specification of surgical utilization, then the data sources and estimation method. Results for equilibrium and disequilibrium assumptions are presented next, concluding with a policy discussion.

2. Neoclassical and Inducement Models

The positive correlation of utilization rates with physician-population ratios has often been cited as evidence that physicians are able to generate demand for their services. Yet, this association can also be explained under traditional economic theory as a market response to increased competition. When the supply of physicians increases with fixed demand, workloads fall as a fixed patient load is spread across more physicians. Physician incomes consequently fall as well. Physicians hypothetically respond to greater competition by lowering their fees, and the quantity demanded of physician services expands at this new (lower) price to clear the market. Note that the original demand curve is not altered in any way.[2]

Inducement theorists like Reinhardt (1978), Evans (1974), and Fuchs (1978)[3] argue that two key characteristics of the competitive marketplace are not present in the medical market, *viz.*, the independence of demand and supply and consumer sovereignty. Restrictions on professional advertising and lack of competition limit the consumer's ability to shop for price when seeking physician services. In addition, the technological complexity of medical care makes it difficult for the consumer to evaluate these services. In his role as consumer agent, the physician defines the patient's medical care needs and recommends treatment. At the same time, however, he also provides the services to meet those needs, which suggests a conflict of interests. The large growth in health insurance coverage merely reinforces any shift incentive, by reducing the net price of care.[4]

Under the inducement hypothesis, market equilibrium can be obtained, not only by adjustments in price, but also by influencing the consumer's perception of need. Physicians respond to more competition by generating greater demand for their services, either by increasing output, raising fees,

or both. Physician incomes (and possibly workloads as well) are likely to fall, but by a smaller amount than in the absence of shifting because fees have fallen less and output increased more than would have occurred in a perfectly competitive market.

Reinhardt (1978) has compared the impacts of increased supply on utilization rates, workloads, fees, and physician incomes under the neoclassical and inducement assumptions. Only physicians' fees offer a definitive test of the inducement hypothesis, for either a zero or positive physician supply coefficent in a fee equation is incompatible with neoclassical theory, assuming visit content is adequately controlled for.

3. The Empirical Model

3.1 System of Equations

A test of the two competing hypotheses was made, first, using a simplified equilibrium model of the physician market:

$SR = d(p, SG, Y)$	: per capita surgery demand	(1)
$W = w(p, SG, N)$	: surgeon workloads	(2)
$SS = W.SG$	: surgery supply per capita	(3)
$SP = s(p, W, A, N)$	: surgeon location	(4)
$SG = SS$	: equilibrium condition	(5)

where SR = per capita surgery rate; p = average surgical fees; SG = surgeon-population ratio; Y = vector of relevant demand characteristics; W = surgeon workloads (i.e., average operations per surgeon), SS = surgery supply per capita; N = vector of cost and productivity influencing variables; and A = vector of professional and community amenities. Solving (5) for the

equilibrium price (p) we have

$$d(\bar{p}, SG, Y) = w(\bar{p}, SG, Y, N).SP, \quad (5a)$$

or in terms of the standard reduced-form price equation

$$\bar{p} = e(SG, Y, N) \quad (5b)$$

Equations (1 - 4) and (5b) represent four structural equations and an identity, with five endogenous variables and several exogenous ones. The target income, or inducement, theory presumes physicians have some discretionary influence over demand; hence the unorthodox inclusion of the surgeon-population ratio in the demand equation. A significant, positive coefficient for SG in eq. (1) would corroborate earlier work by Fuchs (1978) that surgeons can, and systematically do, shift demand when faced with increasing competition.

Because demand and supply may be particularly inelastic, demand shift may manifest itself not so much in increased utilization as in higher fees. Two alternative tests are therefore conducted; one involves estimating the structural demand equation with surgery rates on the left-hand-side while the second estimates the equilibrium price equation (5b), incorporating both (hypothesized) positive inducement and negative supply effects. Positive supply coefficients in (5b) are inconsistent with neoclassical theory, _ceteris paribus_.

3.2 Endogenous Variables

Endogenous variables include surgery rates,[5] surgeon-population ratios, workloads, and average surgeon fees. Surgery rates were specified in three alternative ways: total (SR), nonelective (SRN), and elective (SRE) surgical operations per 1,000 population (obstetric procedures were excluded since they are dependent on conception and birth rates and not of analytic interest.)

Elective procedures are those meeting three criteria: non-emergency, non-life threatening, and unrelated to life expectancy (e.g., cholecystectomy, tonsillectomy, varicose vein excision, and hemmorhoidectomy).[6] Surgeons are hypothesized to have greater discretionary influence over elective procedures, and thus the rates of these procedures are expected to be more sensitive to inducement than operations for life-threatening conditions. The relative supply of surgeons in a market area was measured by the surgeon-population ratio (SG), or the number of office-based surgical specialists per 1,000 population. Surgical workloads were defined as average annual number of operations per surgeon. The gross price (P) of surgical services was measured by a weighted surgical fee index for six common procedures using national operation frequencies as weights to avoid confounding fees with surgical mix.[7] A county-specific cost-of-living deflator based on BLS data was then used to adjust for broad differences in prices. (The equation used to produce the deflator is available upon request.)

3.3 Exogenous Variables

Exogenous variables are represented by the Y, N, and A vectors in the structural equations. Variable definitions, means, and sources are presented in Table 1. All statistics are population weighted. The Y vector consists of demand variables, including sociodemographic characteristics of the population, health status, ability to pay, the price of hospital care, and substitute sources of care.

Socio-demographic characteristics included age, sex, race and education. The health status of the population was measured by the number of reported bed-disability days over two weeks.

TABLE 1

VARIABLE DEFINITIONS, SOURCES, AND MEANS

Variable	Description	Source	Mean
SR	total operations per 1,000 population	HIS	62.18
SRN, SRE	nonelective, elective operations per 1,000 population	HIS	31.23, 30.95
SG	surgeons per 1,000 population	AMA, *Physicians Distribution in the U.S.*	0.30
W	annual operative workload	$\frac{SR}{SP}$	280.76
P	surgical fee index	Medicare prevailing charge data*	323.39
OVER65	proportion 65 years and older	HIS	0.10
AGE4564	proportion aged 45-64 years	HIS	0.21
AGE2144	proportion aged 21-44 years	HIS	0.44
FEM	proportion female	HIS	0.52
WHITE	proportion white	HIS	0.87
HISCH	proportion 9-12 years education (household heads)	HIS	0.47
UNIVSCH	proportion 13 years education and more (household heads)	HIS	0.28
DISDAY	mean bed disability days	HIS	0.25
EMPLOY	proportion employed	HIS	0.58
PROFES	professional and technical workers, managers (proportion of labor force)	HIS	0.26
CLERK	clerical, sales, and service workers (proportion of labor force)	HIS	0.35
BLUCOL	craftsmen, operatives, and laborers (proportion of labor force)	HIS	0.34

TABLE 1 (continued)

VARIABLE DEFINITIONS, SOURCES, AND MEANS

Variable	Description	Source	Mean
INCOME	mean family income (in 1,000s)	HIS	10.7
COIN	proportion of hospital expenses covered by third party payor	predicted from state insurance regression	0.84
HOSP$	hospital expenses per admission in (100s)	American Hospital Guide issue tapes	10.85
GPPT	general practitioners per 1,000 population	AMA, Physician Distribution in the U.S.	0.23
NEARSG	surgeons in adjacent SMSA per 1,000 non-SMSA plus adjacent SMSA population	AMA, Physician Distribution in the U.S.	0.04
WAGE	hourly wage for employees in retail trade	U.S. Census, Census of Retail Trade, 1967, 1972 1977	2.18
DEGDAYS	degree days (in thousands)	Statistical Abstracts	4.46
HOTEL$	hotel receipts (in thousands) per capita	U.S. Census, Economic Surveys-Selected Service Industries, 1967, 1972, 1977	0.50
SMSAPOP	proportion residing in SMSA	HIS	0.67
HBEDPT	short-term general hospital beds per 1,000 population	American Hospital Assoc. Guide issue tapes	4.55
COLLEGE	number of colleges and universities	1969: College Blue Book 1970-76: Chronicle of Higher Education	9.01

*U.S. Senate Subcommittee on Finance, Medicare and Medicaid: Problems, Issues and Alternatives (1969). Personal Communications from Medicare Part B carriers (1970-1973). HCFA Medicare Directory of Prevailing Charges (1974-76).

Two variables measure ability to pay: mean family income (INCOME) and health insurance coverage (COIN), the latter defined as the proportion of hospital expenses covered by all third party payors. Unfortunately, reliable health insurance data were not available at the substate level. Instead, the COIN variable was derived for each physician market from a state insurance regression (available upon request).

Higher prices of hospitalization and patients' time, besides physicians' fees, should discourage utilization [Acton (1975); Grossman (1972)]. The gross price of hospitalization (HOSP$) was measured by the mean hospital expense per admission. With low coinsurance rates, the time price of surgery becomes more important in deciding to undertake elective surgery. Market area occupation distribution and employment status were used as proxies for time price. Employment (EMPLOY) was defined as the proportion of the population who were working at the time of the interview. Three occupational variables were specified: the proportion of the labor force composed of professional and technical workers and managers (PROFES); the proportion clerical, sales, and service workers (CLERK); and the proportion craftsmen, operatives, and laborers (BLUCOL). Farm workers and farm managers constitute the omitted group. Grossman (1972) argues that higher paid white collar workers value time more and hence should have lower utilization, *ceteris paribus*.

The relative supply of general practitioners constitutes one of two potential sources of alternative care affecting demand, and was defined as the number of GPs per 1,000 population (GPPT). In those non-metropolitan areas where migration, or bordercrossing, for care appeared likely, an additional surgeon variable was specified. Nearby SMSA surgeons (NEARSG) was defined as the number of surgeons in the adjacent SMSA per 1,000 SMSA-plus-rural population (see Fuchs, 1978).

The N vector was limited to two complementary input variables: hospital beds and nonphysician wages. Beds per 1,000 (HBEDPT) was defined as the number of short-term general hospital beds per 1,000 population and should increase the supply of operations by improving access, shortening queues, and encouraging surgeons to fill beds (the so-called Roemer Effect). Surgeons are also hypothesized to restrict supply, or require higher fees, where the cost of labor inputs is higher, as measured by retail wage rates, WAGE [Benham *et al*. (1968); Sloan (1974)].

Concerning the location equation, surgeons are expected to practice in areas where they can maximize professional and residential amenities, *inter alia*. Professional amenities included two measures: urbanization (SMSAPOP) and proximity to a medical school. Proximity to a medical school was highly colinear with the presence of other colleges generally, however, and as the latter was a superior predictor, COLLEGE was used instead.

Four variables measured the desirability of the area as a place to live: racial mix of the community (WHITE), climate, cultural opportunities, and its attractiveness as a vacation spot. The climate variable (DEGDAYS) was defined as annual degree days, or the number of days the temperature falls below 65^{o}F times the number of degrees below 65, and should be negatively related to SG. Cultural opportunities were assumed to be provided by the presence of local colleges and universities. Finally, areas with high hotel and motel volume were assumed to be attractive vacation spots; (again suggested by Fuchs, 1978); HOTEL$ was specified as hotel receipts per capita.

Finally, a linear time trend was included to capture secular trends in the demand for, or supply of, surgical services (TIME). Patient attitudes toward surgery may change systematically over time, for example, or innovations in medical technology may alter accepted surgical practice.

4. Data Sources and Estimation Methods

4.1 Data

The Health Interview Surveys (HIS) from 1969 through 1976 were the major data base for this project. The HIS is conducted annually by the National Center for Health Statistics and represents a probability sample of all U.S. households in 349-360 Primary Sampling Units (PSUs), usually SMSAs or collections of rural counties. It includes civilian non-institutionalized household residents who are alive at the time of the interview.[8] Since the HIS is a probability sample, reported utilization of services could be used to develop population-based rates at the micro-area level. This permitted a much finer micro-analysis of variations in utilization than has been possible with data aggregated to the state or regional level (as in Fuchs' research).

However, to the extent patients cross small-area geographic boundaries in search of surgical services, positive correlations between utilization rates and supply will be spurious. As HIS surgery rates are population-based, in-migration will not bias these rates upwards. Patients from rural areas or other SMSAs, however, may cross into an adjacent SMSA area to seek surgical services. Contiguous SMSAs were found only in four instances and were merged and treated as a single PSU for analytic purposes. Bordercrossing to urban areas appeared likely in over one-half of the non-metropolitan PSUs in the 1969-1976 time-series. In these cases, we included an additional surgeon supply variable, NSP, as a control for supply.

Besides surgery, the HIS included the following socio-demographic and health status variables for each member of the household: age; sex; race; highest level of education attained for all members 17 years and over; total family income; occupation and employment status; and bed disability days.

A PSU surgical fee index was based on Medicare prevailing charges for six procedures: cholecystectomy, hernia repair, hemmorhoidectomy, lens extraction, prostatectomy, and radical mastectomy. The prevailing charge is the "maximum allowable payment" set by Medicare and is currently established at the 75th percentile of usual and customary charges in a given geographic area. Fee data for 1974-1976 were obtained from the annual Medicare Directory of Prevailing Charges, and for 1969 from a U.S. Senate Finance Committee report. Published Medicare charge data were not available for 1970-1973, and were obtained directly from the Medicare Part B carriers by personal survey.

4.2 Estimation Method

The structural model was estimated in linear form using two-stage least squares on a sample of 349-360 PSUs over the eight year period, 1969-76, for a pooled sample of 2,837. Because of the presence of simultaneous equation bias, all endogenous variables, SR, P, SG, and W, were first regressed on the complete list of exogenous variables as instruments. Their imputed values were then used in estimating structural coefficients in the second stage.[9] All regressions were weighted using PSU population weights, and dollar variables deflated by an imputed area cost-of-living index.

5. Equilibrium Results

5.1 National Estimates

Regression results for the entire sample are shown in Table 2. Equations (1-3) represent per capita utilization demand equations for all (SR), nonelective (SRN), and elective (SRE) surgery, respectively, while equation (4), the $\bar{p}$ equation, determines equilibrium fees with surgeon density the only endogenous variable.

TABLE 2

TWO-STAGE LEAST SQUARES REGRESSIONS: ENTIRE PSU SAMPLE, 1969-1976
(t-values in parentheses)

INDEPENDENT VARIABLE	DEPENDENT VARIABLE SR (1)	SRN (2)	SRF (3)	P (4)
P	-0.026	-0.010	-0.016	--
	(2.1)**	(1.09)	(2.0)**	
SP	17.53	4.50	13.01	972.7
	(2.1)**	(0.8)	(2.4)**	(14.6)**
W	--	--	--	--
SG	--	--	--	--
OVER65	46.5	43.2	3.3	-255.9
	(3.4)**	(4.4)**	(0.4)	(4.0)**
AGE4564	20.4	26.5	-6.1	-363.0
	(1.6)*	(3.0)**	(0.7)	(6.0)**
AGE2144	12.2	8.3	3.9	-214.8
	(0.9)	(0.9)	(0.4)	(3.4)**
FEM	-5.1	-23.6	18.5	-35.1
	(0.4)	(2.3)**	(2.0)**	(0.5)
WHITE	10.9	2.3	8.6	-83.3
	(3.1)**	(0.9)	(3.8)**	(5.4)**
HISCH	21.6	10.2	11.3	-250.2
	(3.8)**	(2.6)**	(3.1)**	(9.5)**
UNIVSCH	0.3	4.7	-4.4	-370.3
	(0.0)	(0.9)	(0.9)	(9.0)**
DISDAY	20.3	14.5	5.8	43.31
	(5.8)**	(5.9)**	(2.6)**	(2.7)**
INCOME (000)	0.70	0.49	0.21	9.1
	(3.3)**	(3.3)**	(1.5)	(11.4)**
COIN	0.40	0.22	0.17	57.5
	(0.2)	(0.2)	(0.2)	(7.3)**
HOSP$ (00)	-0.22	-0.15	-0.07	0.78
	(6.4)**	(6.2)**	(3.2)**	(5.2)**
EMPLOY	-1.9	3.4	-5.3	-209.0
	(0.3)	(0.7)	(1.1)	(6.6)**
PROFES	-2.4	-4.0	1.6	-437.4
	(0.2)	(0.6)	(0.2)	(8.9)**
CLERK	10.5	2.4	8.1	-488.5
	(1.2)	(0.4)	(1.4)	(10.7)**
BLUCOL	7.8	1.3	6.5	-372.2
	(1.2)	(0.3)	(1.4)	(10.1)**
GPPT	-13.6	-6.5	-7.2	349.2
	(2.4)**	(1.6)	(1.9)**	(12.6)**
NEARSG	2.3	-0.7	3.0	175.8
	(0.6)	(0.2)	(1.2)	(8.5)**
HBEDPT	--	--	--	-22.5
	--	--	--	(13.8)*
WAGE	--	--	--	-0.2
				(0.1)
TIME	0.40	0.58	-0.19	4.41
	(2.2)**	(4.6)**	(1.6)	(4.9)**
CONST	20.1	13.0	7.1	899.8
	(1.4)	(1.3)	(0.8)	(13.1)**
R^2	.09	.08	.05	.22
	(14.0)	(13.0)	(7.6)	(36.9)

*Significant at ten percent level.

**Significant at five percent level.

Both the total and elective surgery demand equations show statistically significant, negative price effects, although the own price elasticities are quite small (-0.14 and -0.17, respectively). Nonelective surgery also shows a negative price effect, albeit insignificant. Fuchs (1978) was unable to verify a price effect for surgery, possibly because he had price data only on office and hospital visits, not operations, and only for 22 broad metropolitan-nonmetropolitan census areas.

Higher gross hospital prices per admission, HOSP$, like physician fees, discourage elective and nonelective surgery as expected, but the cross-price elasticity is also quite small (-.04). Occupational measures of time price were generally insignificant in the demand equation, possibly due to offsetting effects (viz., higher time prices confounded with greater disability insurance, more sick leave, and health status).

Hospital coinsurance rates were found to be significant in the utilization equations, most of their effect coming in terms of prices as discussed below. Hospitalization coverage is now quite extensive and shows very little variation (its coefficient of variation (CV) was .36) in comparison to variations in gross admission charges (CV = 1.1). The relative lack of variation is probably the main reason for its insignificant marginal effect on demand.

Other significant variables in the demand equations include health status (DISDAY, +), family income (INCOME, +), time (TIME, +), middle aged and elderly per capita (AGE4564, OVER65, +), high school graduates (HISCH, +), percent white (WHITE, +), and general practitioners per 1,000 (GPPT, -).

Holding fees and other relevant demand factors constant, surgeon density (SG), derived from a first-stage equation, is significant and positive in both the total and elective (but not the nonelective) demand equations. A 10

percent increase in surgeons per capita (around the mean of .3) results in a 0.9 percent increase in overall surgery per capita and 1.3 percent increase in elective surgery. These elasticities are about one-third those found by Fuchs (1978, p. 47). The positive SP elasticity, due primarily to elective surgery, supports the hypothesis that more discretionary surgical procedures are performed in areas of high surgeon concentration.

Equation (4) represents a long-run equilibrium solution to the model where all explanatory variables, save imputed surgeon density, are exogenous. It is the "price equation" usually referred to in the literature. All demand-specific variables should exhibit positive signs while those entering the supply equation positively should depress prices.

Some variables do show consistent signs (e.g., health status, income, coinsurance, time), but many others do not. Part of the problem is due to the collinearity among predicted SP taken from the first stage and the included variables; thus, the results should be interpreted cautiously. (The possibility that disequilibrium fees are being observed instead is considered below.)

Certainly the most dramatic finding is the strong, positive influence surgeon density has on equilibrium surgical fees. This is true even after purging SG of the feedback effect of higher fees on surgeon location. If the coefficient for SG in eq. (4) is attributed solely to physician inducement, then a 10 percent increase in surgeon density results in fee inducement of 9 percent per operation, or a long-run elasticity of .90. This would imply only a marginal decline in net incomes from surgery with increased competition.[10]

5.2 Regional Estimates

Large cross-sectional variation exists in the types of PSUs being compared, and Ramsey (1980) has warned against specification biases that could result from comparing very disparate markets.[11] To determine the effect of PSU size on our results, the model was re-estimated for three subsamples: nonmetropolitan PSUs; small SMSAs; and large SMSAs. Large SMSAs were defined as those having populations greater than 500,000. Condensed regression results, focussing on the endogenous variables, are presented in Table 3. (For regional variation by census division and surgeon-population quartiles, the reader is referred to Mitchell and Cromwell, 1982).

Summarizing the results, first for non-metropolitan areas, we find the kind of market of neoclassicist would expect: (1) a small negative price elasticity of demand (-.22); (2) no discernable surgical inducement; and (3) falling equilibrium fees with greater surgical concentration. Small metropolitan areas show a different picture in one major respect: higher equilibrium fees with increasing surgeon concentration.

Large metropolitan areas present a dramatic contrast from rural and small metropolitan areas. First, demand--particularly for elective surgery--varies inversely with surgeon fees (e = -.20 for total surgery; e = -.34 for elective surgery).[12] Second, utilization varies positively with surgeon concentration (although the relationship is not statistically significant).[13] And third, equilibrium fees are also positively related to SP.

5.3 Increasing Monopoly Model

Pauly and Satterthwaite (1980) at Northwestern University have proposed a new theory of "increasing monopoly" to account for the positive correlation between physicians per capita and fees. It is predicated on the assumption

TABLE 3

TWO-STAGE LEAST SQUARES REGRESSIONS: NON-METRO, SMALL AND LARGE METRO PSUs, 1969-1976
(t-values in parentheses)

INDEPENDENT VARIABLE	DEPENDENT VARIABLE SR (5)	SRE (6)	$\bar{P}$ (7)
REGION: NON-METRO (1604 obs.)			
P	-0.043 (1.7)*	-0.017 (1.0)	
SG	5.5 (.6)	3.6 (.6)	-215.9 (3.2)**
R^2 (F)	0.08 (7)	0.06 (5)	0.21 (21)
REGION: SMALL METRO (686 obs.)			
P	0.004 (.1)	0.013 (.6)	
SG	5.57 (.2)	12.7 (.8)	459.1 (4.4)**
R^2 (F)	0.06 (2)	0.03 (1.1)	0.22 (10)
REGION: LARGE METRO (546 obs.)			
P	-0.037 (1.7)*	-0.032 (2.2)**	
SG	25.3 (1.2)	14.20 (1.0)	680.0 (6.7)**
R^2 (F)	0.23 (8)	0.14 (5)	0.50 (27)

Note: Complete regression results are available from the authors.

*Significant at ten percent level.
**Significant at five percent level.

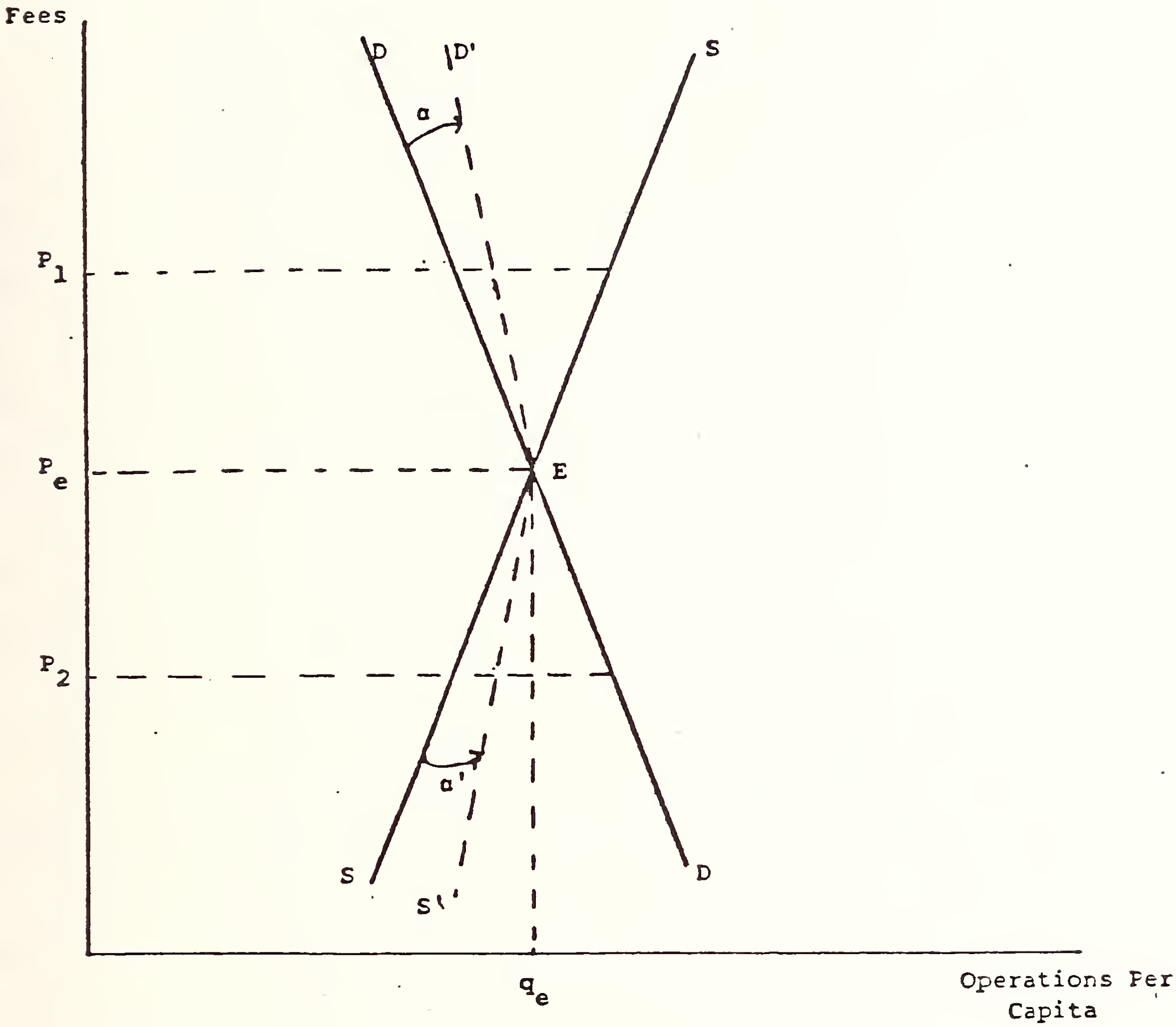
that "consumer information, and consequently the degree of certainty about other sellers' quality levels, may decrease as the number of sellers increases" (p.27). Empirical tests of the hypothesis have centered on alternative specifications of the extent and quality of consumer information. Primary among these is the absolute number of physicians in the consumer's market area, or physicians per square mile.

A limited test of the increasing monopoly model was made by including surgeons per square mile along with surgeons per capita as explanatory variables.[14] (Other proxies for information flows were unavailable.) The inclusion of surgeons per square mile, if anything, increased the significance of the surgeon per capita variables in the structural demand equations, while having a negative, significant impact on equilibrium fees which is counter to the increasing monopoly model. It is possible that other measures of consumer information are needed to fully test the theory. Nevertheless, our results--at least in terms of surgery--appear robust with regard to physician density.

6. Disequilibrium Results

The assumption that fees adjust within the period to equate the demand for and supply of surgical operations is a particularly strong one in light of published statistics on surgeons showing light workloads [ACS (1975); Hughes (1973)]. Surgical fees in many areas may be rigid downwards because insurers key on stated charges in determining allowables, with desired supply perpetually exceeding desired demand.[15] Hence, observed utilization figures should be on, or very near, the demand curve, DD, in Figure 1, presuming that "shortside" demanders-of-care realize their planned trade, while suppliers' preferences remain unfulfilled [Green (1978) p. 30]. The inducement

FIGURE 1: A Disequilibrium Inducement Model



hypothesis contends, however, that surgeons in excess supply can and do exert some discretionary power over demand, effectively rotating it to the right by some (unknown) degree, α , to D', for example.

Of course, many but certainly not all surgeons practice in glutted markets, and Yett (1978) argues that the positive association between fees and utilization may be due to a small, but important, number of physicians practicing in underserved areas, partially filling an excess demand gap. If we are observing price-quantity combinations not along demand, DED, but DES', then a positive correlation between fees and surgeon supply is, in fact, possible, without any real inducement. The observed correlation could be due to a specification bias arising from chronic disequilibrium.

Green (1978) points out that if the observations can be separated into excess supply and demand regimes, then a direct estimate of the shift effect could be derived from estimating

$$Q = \begin{cases} S + \alpha' (D-S): & D > S \\ D + \alpha (S-D): & D < S \end{cases} \qquad (6)$$

and evaluating the supply coefficients (including surgeons per capita) where $D < S$. Without inducement, should equal zero; that is, observed surgery rates should be a function of demand factors alone in excess supply areas.

Expanding (6) we have

$$SR = (1 - \alpha')S + \alpha'D: \quad D > S \qquad (6a)$$

and

$$SR = (1 - \alpha)D + \alpha S: \quad D < S \qquad (6b)$$

Substituting the respective demand and supply vectors (including P and SG) for D and S and estimating (6a) and (6b) using OLS, we should find that in excess supply regions utilization is demand-dominated while in excess demand regions supply is more constraining. Also, by segmenting our sample in this way and

focussing on excess supply regions, there is less chance that a positive inducement effect may simply be due to physician shortages in a few areas. If demand shift is found in surplus areas, this would reinforce the earlier, equilibrium findings.

Segmenting the sample is difficult to do a priori because we do not know desired demands and supplies at going market fees for individual areas. As a first approximation we split the PSU sample at the mean physician workload of 280 operations per surgeon. Those PSUs whose average workloads were above 280 were considered "shortage" areas while those where $W < 280$ were designated "surplus" areas.[16] Two hundred thirty-three PSUs were considered surgeon-surplus areas using this algorithm; 119, shortage areas. Certainly some classification error exists in such a procedure, but the results should be less contaminated than under the pooling assumption of single period equilibrium.

The results are reported in Table 4. As expected for those physicians practicing in "shortage" areas, observed utilization rates are almost exclusively determined by physician supplies (SG); fees and most socio-economic factors have no discernable effect. Only disability days and education show a statistically significant positive effect on utilization. For surgeon-scarce areas, the surgery rate elasticity with respect to surgeon concentration is .28 compared with .09 for the entire sample.

Where surgeons appear abundant, demand variables play a much greater role. Fees show a significant, negative effect on utilization ($e = -.13$); income a positive effect ($e = .09$). (The negative sign for the coinsurance variable is theoretically unexpected, however.) Other demand-side variables like age distribution, disability days, education, and the average cost of a hospital admission ($e = -.04$) were also significant.

TABLE 4

TOTAL SURGERY RATE OLS REGRESSIONS FOR HIGH AND LOW SURGEON WORKLOAD AREAS (t-values in parentheses)

INDEPENDENT VARIABLE	TYPE OF MARKET Shortage (8)	Surplus (9)
P	-0.009 (0.5)	-0.023 (3.0)**
SG	132.3 (12.5)**	13.26 (2.8)**
OVER65	34.2 (1.6)	62.5 (4.2)**
AGE4564	25.5 (1.3)	27.0 (0.2)
AGE2144	27.8 (1.3)	3.5 (0.2)
FEM	-2.11 (0.1)	-19.3 (1.2)
WHITE	6.7 (1.0)	5.0 (1.3)
HISCH	19.5 (2.3)**	15.0 (2.4)**
UNIVSCH	-5.4 (0.5)	5.6 (0.8)
DISDAY	13.4 (2.6)**	16.6 (4.2)**
INCOME (000)	0.6 (1.6)	0.5 (2.8)**
COIN	4.5 (1.1)	-3.59 (1.7)*
HOSP$ (00)	0.5 (1.4)	-0.2 (5.5)**
EMPLOY	-16.0 (1.5)	9.3 (1.2)
PROFES	-13.7 (0.9)	-0.3 (0.0)
CLERK	10.8 (0.8)	3.8 (0.4)
BLUCOL	-4.6 (0.4)	-5.5 (0.7)
GPPT	-8.5 (1.0)	-14.4 (2.4)**
NEARSG	0.4 (0.1)	-6.3 (1.2)
HBEDPT	75.5 (0.2)	33.8 (0.1)
WAGE	0.19 (0.1)	0.18 (0.4)
SMSA	3.1 (1.0)	5.0 (3.1)**
TIME	0.47 (1.4)	0.18 (1.0)
CONST	12.3 (0.6)	35.1 (2.5)**
R^2 (F)	0.35 (19.6)	0.16 (13.2)

*Significant at ten percent level.
**Significant at five percent level.

Even in excess supply areas, however, the impact of physician concentration on utilization does not disappear. Surgeon availability continues to have a significant, positive sign although the elasticity has been reduced (.08). Furthermore, an inelastic demand implies that most of the inducement effect is reflected in higher fees and not operations *per se*.

7. Policy Implications

The 1975 study of surgical manpower prepared by the American College of Surgeons (1975) found the current surgical manpower pool in the United States to be more than adequate, at least in terms of quantities if not quality and specialty mix. General and family practitioners in particular were encouraged to redirect their efforts away from surgery. The surgeons also argued against regulating fees too severely in order to preserve returns on the extensive human capital investments made by their colleagues.

Regulators may legitimately question the ultimate impacts on competition of such restrictions in supply. With very price-inelastic demand, reductions in surgical residencies and foreign-trained immigrants may enable practicing surgeons to charge much higher fees and enjoy monopolistic incomes far beyond those that would obtain from freer entry.

Our results, furthermore, provide little reassurance that significant additions to the surgical manpower pool will work to keep prices down. Burgeoning supplies necessarily mean declining workloads which, combined with downward rigid fees due to UCR-based insurance and inelastic demand, encourage fee increases and a (likely) reallocation of time to non-surgical activity for which the provider is often overqualified.

Solutions to the problem do not come easily. If demand cannot be considered as fixed, then fee regulation may be met by increases in

discretionary surgery. A similar weakness may be true of higher patient copays, although the higher out-of-pocket prices of induced surgery should discourage it to some extent. On the other hand, if surgery itself is controlled through peer review (including Professional Standards Review Organizations or second opinion surgery requirements), then inflationary pressures on fees are a probable outcome.

Possibly a more equitable and in the long-run more viable approach involves reimbursement schemes which encourage surgeons and their referring colleagues to simultaneously monitor both fees and utilization rates. HMOs are the most obvious examples of this approach although neighborhood clinics, Independent Practice Associations, Preferred Provider Organizations, and the like could have similar incentive structures depending on how physicians are paid and how patients participate.

In addition, now that Medicare is paying hospitals on a fixed charge per case, serious consideration is being given to "physician DRGs" in the hospital. This involves packaging all physician care, surgical and medical, into a single global rate per Diagnostic Related Group. Whether this would discourage surgery is unknown. On the one hand, surgical DRGs are considered more costly and enjoy higher reimbursement rates. But where they put the surgeon at risk for other physician services, they could discourage surgery. Assuming discretionary surgery to be less physician intensive, and hence less costly, perverse results could obtain from DRG payment if within-DRG severity is not adequately controlled for.

FOOTNOTES

1. Studies of prepaid group practices have demonstrated significantly lower rates of surgical utilization compared with traditional fee-for-service practices (Gaus, 1976), but prepaid group practices and health maintenance organizations are still the exception and not the rule in this country.
2. An increase in the number of physicians will also increase utilization by lowering the time price of care. Greater access to physicians presumably reduces travel distances as well as queues in the physician's office. Thus, some indirect demand "shift" is implicit in neoclassical models as well, in the form of lower time prices (see Sloan and Lorant, 1977).
3. See also Feldstein (1970), Dyckman (1978), Redisch, et al. (1977), Davis and Reynolds (1976), Fuchs and Kramer (1972), Lee and Hadley (1981), and Held and Manheim (1980).
4. The net price of physician office care, for example, fell from 56% to 43% of the real office price from 1966-76, even with a 10% increase in real prices (BHPA, 1981, Table 9).
5. Surgical operations are used to test the ability of physicians to shift demand for their services in response to changes in the physician-population ratio because (a) operations are discrete procedures that are homogeneous with respect to definition and duration across the country, (b) it is unlikely that observed surgical utilization is typically supply constrained with fees below equilibrium, which would prevent a direct estimate of the demand curve as only price-quantity points along the supply curve are observed (see Feldstein, 1970), and (c) reduced search and travel costs associated with high physician density are probably not an important factor in the increased demand for operations,

as Fuchs (1978) has noted, given the relatively high psychic costs of surgery and time costs of hospitalization.

6. For more discussion on how diagnostic information was categorized as to elective and nonelective, see Mitchell and Cromwell (1980, Appendix C).

7. Ideally, we would have liked to calculate a net price variable, adjusted by the appropriate coinsurance rate. Unfortunately, coninsurance data for surgical services _per se_ were unavailable. As surgical insurance coverage is almost perfectly correlated with hospitalization coverage (which we did measure), this omission is not a serious problem.

8. For details on the accuracy of HIS data, see NCHS (1965).

9. Auster and Oaxaca (1981) raise an important issue in identifying the inducement effect in the demand equation by pointing out that unless, say, operations are performed under varying physician-nonphysician input combinations, output will be tautologically related to physician supply. Factor price variations, they argue, might not be large enough to assure significantly divergent factor input mixes. Ignored in their discussion, however, are physician "wage rates" which may vary substantially across areas even if auxiliary input prices do not, e.g., surgeons in the South, for example, may prefer to do more-or-less surgery. Also key in surgery is access to hospital beds which in some major hospitals, if not cities, is a real queueing problem for elective surgery.

10. For an interesting insight into why surgeons are not poor when there are so many of them and they do so little surgery, see Owens (1974).

11. While our fee index uses national surgical proportions as weights, thereby purging it of surgery-mix effects across markets, this may only partially adjust for the interarea variation in surgical complications.

12. The cross-price elasticity of hospital costs, HEXPA, was also significantly negative, i.e., -.06.

13. Decomposing the sample by metropolitan area reduces the significance of the relationship between surgery demand and SP. The fact that the SP coefficients are all positive in the SR and SRE equations in Table 3, although insignificant individually, add up to a significant relationship for the combined sample.
14. In Pauly-Satterthwaite's empirical work, these two measures were highly correlated, which is not surprising as their work was confined to the 100 largest SMSAs. Using a national sample of 350 PSUs, urban and rural, reduces the correlation to .29, permitting a finer separation of the two variables.
15. In surveys, surgeons clearly state that they would be willing to perform more operations at prevailing prices (see Hughes (1973)).
16. 280 operations per year works out to 5.5 operations per week for 50 weeks, a number somewhat higher than found in the SOSSUS report (ACS, 1975). Part of the difference is due to operative workloads of nonsurgeons which are included in the numerator of our workload measure. Using workloads as a measure of market disequilibrium also presumes the underlying demand-supply characteristics in shortage and surplus areas to be similar on average, which is debatable.

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